

- 13** In an experiment, a heater supplies heat at a constant rate to a liquid in an insulated container of negligible heat capacity. The temperature of the liquid rises steadily at 6.0 K per minute for 8 minutes after which it begins to boil. After boiling for 15 minutes, all the liquid has vapourised.

What is the ratio of the specific heat capacity to the specific latent heat of vaporization of the liquid?

A $\frac{1}{120} K^{-1}$

B $\frac{1}{90} K^{-1}$

C $\frac{8}{90} K^{-1}$

D $\frac{1}{20} K^{-1}$